

Javascript

Arrays and its methods

Theory

1. Objects in JavaScript

Definition: Objects are collections of key-value pairs used to store structured data. **Use case:** Represent entities with properties and methods. **Key point:** Properties can be added, updated, or deleted dynamically.

Syntax:

```
let person = { name: "FIIT", course: "Javascript" };
```

2. Object Methods

- `Object.freeze()` → prevents modification of properties.
- `Object.assign()` → merges objects.
- **Spread operator (...)** → ES6 shorthand for merging/cloning.
- `Object.keys()` / `Object.values()` / `Object.entries()` → retrieve keys, values, or key-value pairs.

3. Arrays in JavaScript

Definition: Arrays are ordered collections of values. **Use case:** Store lists of data. **Key point:** Arrays can hold mixed data types.

Syntax:

```
let marks = [10, 20, 30];
```

4. Array Methods

- **Mutating methods:** `push()`, `pop()`, `shift()`, `unshift()`, `splice()`.
- **Non-mutating methods:** `slice()`, `concat()`, `map()`, `filter()`, `reduce()`.
- **Search methods:** `indexOf()`, `lastIndexOf()`, `includes()`, `find()`, `findIndex()`.
- **Iteration methods:** `forEach()`, `map()`.

5. Higher-Order Functions

Definition: Functions that take other functions as arguments or return them.

Examples: `map()`, `filter()`, `reduce()`.

Use case: Functional programming style for cleaner, reusable code.

Coding Round Questions

- Object freeze → Show how `Object.freeze()` prevents modification.
- Object merge → Merge two objects using `Object.assign()` and spread operator.
- Array methods → Demonstrate `push`, `pop`, `shift`, and `unshift`.
- Array search → Use `indexOf`, `lastIndexOf`, and `includes` on an array.
- Map and filter → Write a program to double array values using `map()` and filter even numbers using `filter()`.
- Reduce → Use `reduce()` to calculate the sum of an array.

⚡ Machine Coding Round Challenge

Problem: Student Management System

Requirements:

- Create an array of student objects with properties: `name`, `age`, `marks`.
- Write functions to:
 - Add a new student.
 - Find a student by name.
 - Filter students with marks above 50.
 - Calculate average marks using `reduce()`.
- Use `Object.freeze()` to prevent modification of student details once added.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Object</title>

    <style>
      h1 {
        /* border-color: green;
           border-width: 20px;
           border-style: solid; */

        border: 10px solid blue;
      }
    </style>
  </head>
  <body>
    <h1>Object</h1>

    <script>
      let person = {
        name: "FIIT",
        course: "Javascript",
      };

      Object.freeze(person);

      person.name = "kiki";
      console.log(person);

      delete person.name;

      console.log(person);

      console.log(Object.hasOwnProperty("course"));

      let name = "Ganesh";
      let age = 25;

      const student = { name, age };

      console.log(student);

      let obj1 = { name: "Harini", age: 20 };

      let obj2 = { college: "DKM", degree: "BCA" };
```

```
//es6+

//    spread operator
let mergedd = { ...obj1, c: 3 };

console.log(mergedd);

let merged = Object.assign(obj1, obj2);

console.log(merged);

//    arrays

let marks = [10, 20, 30, 40, 50];

console.log(marks);
console.log(marks[0]);
console.log(marks[1]);
console.log(marks[2]);
console.log(marks[3]);
console.log(marks[4]);
console.log(marks.length);

let mixed = [10, 20, "marks", "data", true, false];

for (let i = 0; i < marks.length; i++) {
    console.log(marks[i]);
}

let arr = [10, 20, 30, 40, 50];

// arr.push(60); // add at last
// arr.pop();
// arr.pop();

arr.unshift(80);
arr.shift();

console.log(arr);

let studentName = ["Brindha", "Niha", "Arun", "Niha", "Hema"];

console.log(studentName.indexOf("Niha"));

console.log(studentName.lastIndexOf("Niha"));

console.log(studentName.includes("Raj"));
```

```

console.log(studentName.slice(1, 4)); // n-1 (4-1) => 3

console.log(studentName.splice(1, 2));
console.log(studentName);

let arr1 = [11, 23, 30, 40, 50];

// Map

let newArray = arr1.map((x) => x * 10);

console.log(arr1);
console.log(newArray);

// filter

let newestArray = arr1.filter((x) => x % 2 == 0);

console.log(newestArray);

let result = arr1.find((x) => x % 2 == 0);
console.log(result);

let result1 = arr1.findIndex((x) => x % 2 == 0);
console.log(result1);

arr1.forEach((x) => {
  console.log(x * 2);
});

// reduce

let result2 = arr1.reduce((sum, x) => sum + x, 100);
console.log(result2);

// let a = [1, 2];
// let b = [3, 4];
// let c = a.concat(b); // [1,2,3,4]

// console.log(c);

let a = [1, 2];
let b = [3, 4];

let mergeddd = [...a, ...b];
console.log(mergeddd); // [1,2,3,4]
</script>
</body>

```

```
</html>
```